

Overview of Light Schrödinger Bridge

Let Ω be the state space, e.g., $C([0, 1], \mathbb{R}^d)$, and let $\mathcal{P}(\Omega)$ be the space of all probability measures on Ω . For any measure $S \in \mathcal{P}(\Omega)$ denote its marginal distribution at $t = 0, 1$ as π^S and denote as $T_{|x_0, x_1}$ its bridges on the values x_0, x_1 at times $t = 0, t = 1$ respectively.

Let $W^\epsilon \in \mathcal{P}(\Omega)$ be the Wiener Process in Ω with volatility $\epsilon > 0$, which starts at the prescribed marginal $p_0 \in \mathcal{P}(\mathbb{R}^d)$ at $t = 0$, meaning it satisfies the following SDE:

$$dW_t^\epsilon = \sqrt{\epsilon} dW_t, \quad W_0^\epsilon \sim p_0$$

Let $S^* \in \mathcal{P}(\Omega)$ be the unique solution of the Schrödinger problem (defined by the prescribed marginals $p_0, p_1 \in \mathcal{P}(\mathbb{R}^d)$ and the reference measure W^ϵ). Then it can be disintegrated as:

$$S^*(\cdot) = \int_{\mathbb{R}^d \times \mathbb{R}^d} W_{|x_0, x_1}^\epsilon(\cdot) \pi^{S^*}(dx_0, dx_1)$$

where π^{S^*} is the solution of the EOT problem with quadratic cost, on the space of all admissible plans $\Pi(p_0, p_1)$.

This measure is described by the following SDE:

$$dZ_t = g^*(t, Z_t)dt + dW_t^\epsilon, \quad Z_0 \sim p_0$$

where $g^* : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the optimal drift of the Schrödinger bridge S^* .

Furthermore, the optimal plan can also be written using the Schrödinger potentials $\psi^*, \phi^* : \mathbb{R}^d \rightarrow \mathbb{R}_+$, as:

$$\pi^{S^*}(x_0, x_1) = \psi^*(x_0) \exp \left\{ -\frac{\|x_0 - x_1\|^2}{2\epsilon} \right\} \phi^*(x_1)$$

and the optimal drift can be derived from ϕ^* as:

$$g^*(t, x) = \epsilon \nabla_x \log \int_{\mathbb{R}^d} N(z; x, (1-t)\epsilon \mathbf{I}) \phi^*(dz)$$

The main idea of the Light Schrödinger Bridge is to recover a parametric approximation $\pi_\theta \approx \pi^{S^*}$ which will be used to construct an approximation $S_\theta \approx S^*$ of the SB process ([Light Schrödinger Bridge](#)).

To learn π_θ we need to minimize the KL divergence between π^{S^*} and π_θ :

$$\theta^* = \arg \min_{\theta \in \Theta} \text{KL}(\pi^{S^*} \mid \pi_\theta)$$

Define the adjusted Schrödinger potentials as:

$$u^*(x_0) = \exp \left\{ -\frac{\|x_0\|^2}{2\epsilon} \right\} \psi^*(x_0), \quad v^*(x_0) = \exp \left\{ -\frac{\|x_1\|^2}{2\epsilon} \right\} \phi^*(x_1)$$

And using them, the optimal coupling can be written as:

$$\pi^{S^*}(x_0, x_1) = u^*(x_0) \exp \{ \langle x_0, x_1 \rangle / \epsilon \} v^*(x_1)$$

which means that $\pi^{S^*}(x_1 | x_0) \propto \exp \{ \langle x_0, x_1 \rangle / \epsilon \} v^*(x_1)$.

We define the parameterization of $v^*(x_1)$ as $v_\theta(x_1)$ and with that we can define the parameterized coupling $\pi_\theta(x_0, x_1)$ from v_θ :

$$\begin{aligned} \pi_\theta(x_0, x_1) &= p_0(x_0) \pi_\theta(x_1 | x_0) \\ &= p_0(x_0) \frac{\exp \{ \langle x_0, x_1 \rangle / \epsilon \} v_\theta(x_1)}{c_\theta(x_0)} \end{aligned}$$

where c_θ is the normalization $c_\theta(x_0) = \int_{\mathbb{R}^d} \exp \{ \langle x_0, x_1 \rangle / \epsilon \} v_\theta(dx_1)$.

In the original paper [Light Schrödinger Bridge](#), they proved that for that parameterization it holds that the KL objective admits the representation:

$$\text{KL}(\pi^{S^*} | \pi_\theta) = L(\theta) - L^*$$

where $L^* \in \mathbb{R}$ is a constant depending on p_0 , p_1 , and ϵ , and $L(\theta)$ is given by:

$$L(\theta) = \int_{\mathbb{R}^d} \log c_\theta(x_0) p_0(dx_0) - \int_{\mathbb{R}^d} \log v_\theta(x_1) p_1(dx_1)$$

The author notices that for $x_0 = 0$ $\pi^{S^*} \propto v^*(x_1)$, which means that v^* can be viewed as an unnormalized density. Thus they proposed to define it as a Gaussian Mixture, deriving the parameterization of the Schrödinger potential v_θ :

$$v_\theta(x_1) = \sum_{k=1}^K a_k N(x_1; r_k, \epsilon S_k)$$

where the parameters are $\theta = \{a_k, r_k, S_k\}_{k=1}^K$, with $a_k \geq 0$, $r_k \in \mathbb{R}^d$, and $S_k \in \mathbb{R}^{d \times d}$ positive and symmetric.

Using that parameterization we can derive the conditional coupling:

$$\pi_\theta(x_1 | x_0) = \frac{1}{c_\theta(x_0)} \sum_{k=1}^K \bar{a}_k(x_0) N(x_1; r_k + S_k x_0, \epsilon S_k)$$

where $\bar{a}_k(x_0) = a_k \exp \{ (x_0^\top S_k x_0 + 2r_k^\top x_0) / (2\epsilon) \}$, and $c_\theta(x_0) = \sum_{k=1}^K \bar{a}_k(x_0)$.

Furthermore, let v_θ be an unnormalized Gaussian Mixture and π_θ be given by $p_\theta(x_0, x_1) = p_0(x_0)\pi_\theta(x_1 | x_0)$, then the measure S_θ with $S_{\theta|_{x_0, x_1}} = W_{|x_0, x_1}^\epsilon$, and $\pi^{S_\theta} = \pi_\theta$, is a diffusion process governed by the following SDE:

$$dX_t = g_\theta(t, X_t)dt + \sqrt{\epsilon}dW_t, \quad X_0 \sim p_0$$

where the drift g_θ can be described with the following closed form:

$$g_\theta(t, x) = \epsilon \nabla_x \log \left\{ N(x; 0, \epsilon(1-t)\mathbf{I}) \sum_{k=1}^K [a_k N(r_k; 0, \epsilon S_k) N(h_k(t, x); 0, A_k^t)] \right\}$$

where the quantities $h_k(t, x)$, and A_k^t are given by:

$$h_k(t, x) = \frac{1}{\epsilon(1-t)}x + \frac{1}{\epsilon}S_k^{-1}r_k$$

$$A_k^t = \frac{t}{\epsilon(1-t)}\mathbf{I} + \frac{1}{\epsilon}S_k^{-1}$$

The last results the authors of [Light Schrödinger Bridge](#) proved is the universal approximation property of the Gaussian Mixture parameterization:

Assume that the prescribed marginals p_0, p_1 are compactly supported. Then for all $\delta > 0$ there exists a Gaussian mixture v_θ with scalar covariances $S_k = \lambda_k \mathbf{I} > 0$, that satisfies the inequality:

$$\text{KL}(S^* | S_\theta) = \text{KL}(\pi^{S^*} | \pi_\theta) < \delta$$

Overview of Optimal Schrödinger Bridge Matching

For every coupling $\pi \in \Pi(p_0, p_1)$ let T_π be the reciprocal process of π , i.e.,

$$T_\pi = \int_{\mathbb{R}^d \times \mathbb{R}^d} W_{|x_0, x_1}^\epsilon \pi(dx_0, dx_1)$$

Define, also, the reciprocal projection of an arbitrary measure $T \in \mathcal{P}(\Omega)$ as:

$$\text{proj}_{\mathcal{R}}(T) = \int_{\mathbb{R}^d \times \mathbb{R}^d} W_{|x_0, x_1}^\epsilon \pi^T(dx_0, dx_1)$$

For any reciprocal process T_π , its Markovian projection $\text{proj}_{\mathcal{M}}(T_\pi)$ is a diffusion process defined by the following SDE that preserves all time marginals of T_π ([Light and Optimal Schrödinger Bridge Matching](#)):

$$dX_t = g(t, X_t)dt + \sqrt{\epsilon}dW_t$$

The drift is analytically given by:

$$g(t, x_t) = \int_{\mathbb{R}^d} \frac{x_1 - x_t}{1 - t} p_{T_\pi}(x_1 | x_t) dx_1$$

which is the solution to the following minimization problem:

$$\min_g \left\{ \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(x_0, x_1) \sim \pi, x_t \sim W_{|x_0, x_1}^\epsilon} \left[\left\| g(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right] \right\}$$

Let $\mathcal{S}$ be the space of all Schrödinger Bridges:

$$\mathcal{S} = \{S \in \mathcal{P}(\Omega) : \text{there exist } p_0^S, p_1^S \in \mathcal{P}(\mathbb{R}^d), \text{ such that } S = \arg \min_{T \in \mathcal{F}(p_0^S, p_1^S)} \text{KL}(T | W^\epsilon)\}$$

We define the optimal projection for a reciprocal process T_π as:

$$\text{proj}_{\mathcal{S}}(T_\pi) = \arg \min_{S \in \mathcal{S}} \text{KL}(T_\pi | S)$$

The authors of [Light and Optimal Schrödinger Bridge Matching](#) proved that for a given map $\pi \in \Pi(p_0, p_1)$ the optimal projection of a reciprocal process T_π leads to the Schrödinger Bridge S^* of p_0, p_1 :

$$\text{proj}_{\mathcal{S}}(T_\pi) = S^*$$

Let $\mathcal{S}(p_0) \subset \mathcal{S}$ be the subset of all Schrödinger Bridges that start at the marginal $p_0 \in \mathcal{P}(\mathbb{R}^d)$ at time $t = 0$. Since the Schrödinger Bridge we aim to approximate starts at p_0 at time $t = 0$, i.e., $S^* \in \mathcal{S}(p_0)$ then it suffices to optimize over $\mathcal{S}(p_0)$.

Let $S \in \mathcal{S}(p_0)$ be a Schrödinger Bridge starting at p_0 at time $t = 0$, and let $\pi \in \Pi(p_0, p_1)$ be admissible transport plan between p_0 , and p_1 . Then the optimal projection objective of its reciprocal process is derived as:

$$\text{KL}(T | S) = C(\pi) + \frac{1}{2\epsilon} \mathbb{E}_{t \sim U[0,1], (x_0, x_1) \sim \pi, x_t \sim W_{|x_0, x_1}^\epsilon} \left[\left\| g_S(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right]$$

where g_S is the Schrödinger drift of S , and $C(\pi)$ a quantity independent of S . Therefore the Schrödinger Bridge S^* is given as the solution to the following minimization problem:

$$S^* = \arg \min_{S \in \mathcal{S}(p_0)} \left\{ \frac{1}{2\epsilon} \mathbb{E}_{t \sim U[0,1], (x_0, x_1) \sim \pi, x_t \sim W_{|x_0, x_1}^\epsilon} \left[\left\| g_S(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right] \right\}$$

Because every process $S \in \mathcal{S}(p_0)$ is uniquely determined by its terminal adjusted potential $v : \mathbb{R}^d \rightarrow \mathbb{R}_+$, and every valid potential v uniquely characterizes a velocity drift g_v , optimizing over the infinite-dimensional space of processes $\mathcal{S}(p_0)$ is strictly equivalent to optimizing over the space of potentials:

$$v^* = \arg \min_{v \in \mathcal{V}} \left\{ \frac{1}{2\epsilon} \mathbb{E}_{t \sim U[0,1], (x_0, x_1) \sim \pi, x_t \sim W_{|x_0, x_1}^\epsilon} \left[\left\| g_v(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right] \right\}$$

To make this optimization tractable, we restrict the search space to a parametric family of unnormalized Gaussian mixtures $v_\theta(x_1) = \sum_{k=1}^K \alpha_k N(x_1 \mid r_k, \epsilon S_k)$. This transforms the objective into a minimization over the finite-dimensional parameter space $\theta = \{\alpha_k, r_k, S_k\}_{k=1}^K$.

Under this specific Gaussian mixture parameterization, the drift g_θ resolves analytically into the exact closed-form expression:

$$g_\theta(t, x) = \epsilon \nabla_x \log \left\{ N(x \mid 0, \epsilon(1 - t)\mathbf{I}) \sum_{k=1}^K [a_k N(r_k \mid 0, \epsilon S_k) N(h_k(t, x) \mid 0, A_k^t)] \right\}$$

By substituting this exact parametric drift g_θ into the optimal projection framework, it can be shown that minimizing the mean squared error of the closed-form drift is mathematically identical to minimizing the KL divergence $\text{KL}(\pi^{S^*} \parallel \pi_\theta)$.

Specifically, the regression objective evaluates identically to the dual Entropic Optimal Transport divergence up to an additive constant $\bar{C}(\pi)$ that is entirely independent of θ :

$$\frac{1}{2\epsilon} \mathbb{E}_{t \sim U[0,1], (x_0, x_1) \sim \pi, x_t \sim W_{|x_0, x_1}^\epsilon} \left[\left\| g_\theta(x_t, t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right] = \bar{C}(\pi) + \text{KL}(\pi^{S^*} \parallel \pi_\theta)$$

Proposition 1 (Universal Approximation of the Schrödinger Bridge). *Assume the prescribed marginal distributions p_0 and p_1 are compactly supported. Let π^{S^*} be the true optimal plan of the Schrödinger Bridge, and let π_θ be the parameterized plan induced by an unnormalized Gaussian mixture potential with parameter set $\theta \in \Theta_K$ consisting of K components. Then, optimizing over the finite-dimensional parameter space Θ_K asymptotically recovers the global minimum over the infinite-dimensional space of all valid potentials, such that:*

$$\lim_{K \rightarrow \infty} \inf_{\theta \in \Theta_K} \text{KL}(\pi^{S^*} \parallel \pi_\theta) = 0$$

Proof. Let $\mathcal{V}$ denote the infinite-dimensional space of all valid terminal adjusted potentials, and let $\mathcal{V}_\Theta = \bigcup_{K=1}^\infty \mathcal{V}_K$ denote the space of all unnormalized Gaussian mixture

potentials, where $\mathcal{V}_K$ represents the subset of potentials parameterized by exactly K components.

Because $\mathcal{V}_\Theta \subset \mathcal{V}$, to show that these optimizations are asymptotically equivalent, we rely on the universal approximation property of Gaussian mixtures.

Observe that any mixture of K components can be represented as a mixture of $K + 1$ components by setting the weight of the additional component to zero $\mathcal{V}_K \subset \mathcal{V}_{K+1}$. Consequently, the sequence of infima over these spaces is monotonically non-increasing and bounded below by the non-negativity of the KL divergence:

$$\inf_{\theta \in \Theta_K} \text{KL}(\pi^{S^*} \mid \pi_\theta) \geq \inf_{\theta \in \Theta_{K+1}} \text{KL}(\pi^{S^*} \mid \pi_\theta) \geq 0$$

By the universal approximation property of Gaussian mixtures, for any arbitrary $\delta > 0$, there exists some finite number of components K_δ and a specific parameter set $\theta' \in \Theta_{K_\delta}$ such that:

$$\text{KL}(\pi^{S^*} \mid \pi_{\theta'}) < \delta$$

Because the sequence of infima is non-increasing, it follows that for all $K \geq K_\delta$, the infimum over Θ_K is strictly bounded from above by δ :

$$0 \leq \inf_{\theta \in \Theta_K} \text{KL}(\pi^{S^*} \mid \pi_\theta) \leq \text{KL}(\pi^{S^*} \mid \pi_{\theta'}) < \delta$$

Since this sequence of infima is monotonically non-increasing, strictly non-negative, and can be bounded below any arbitrary $\delta > 0$ for a sufficiently large K , it mathematically satisfies the formal definition of a limit converging to zero:

$$\lim_{K \rightarrow \infty} \inf_{\theta \in \Theta_K} \text{KL}(\pi^{S^*} \mid \pi_\theta) = 0$$

□

Algorithm 1: LightSB-M

Input: plan $\pi \in \Pi(p_0, p_1)$ accessible by samples, adjusted Schrödinger potential v_θ parameterized by a Gaussian mixture $\theta = \{a_k, r_k, S_k\}_{k=1}^K$.

while *Not Converged* **do**

Sample batch pairs $\{x_0^n, x_1^n\}_{n=1}^N \sim \pi$

Sample batch $\{t_n\}_{n=1}^N \sim U[0, 1]$

Sample batch $\{x_t^n\}_{n=1}^N \sim W_{|x_0, x_1}^\epsilon$

$$L_\theta \leftarrow \frac{1}{N} \sum_{n=1}^N \left\| g_\theta(t_n, x_t^n) - \frac{x_1^n - x_0^n}{1 - t_n} \right\|^2$$

Update θ using $\partial_\theta L_\theta$

Output: Learned drift g_θ approximating the optimal g^* .

Algorithmic Implementation

The algorithm dictates that the loss is calculated as:

$$L(\theta) = \mathbb{E}_{t \sim U[0,1], (x_0, x_1) \sim \pi \mid x_t \sim W_{|x_0, x_1}^\epsilon} \left[\left\| g_\theta(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right]$$

We can substitute the definition of the Brownian Bridge and write:

$$x_t = tx_1 + (1 - t)x_0 + \sqrt{\epsilon t(1 - t)}z, \quad z \sim N(0, \mathbf{I})$$

Therefore the original objective becomes:

$$L(\theta) = \mathbb{E}_{t \sim U[0,1], (x_0, x_1) \sim \pi \mid x_t \sim W_{|x_0, x_1}^\epsilon} \left[\left\| g_\theta(t, x_t) - (x_1 - x_0) - \sqrt{\frac{\epsilon t}{1 - t}} z \right\|^2 \right], \quad z \sim N(0, \mathbf{I})$$

As $t \rightarrow 1$, the variance of the target diverges to infinity due to the $\sqrt{\frac{\epsilon t}{1 - t}}$ term, leading to severe training instability. To mitigate this without shifting the global minimum of the regression, we introduce a time-dependent weighting function $\lambda(t) = 1 - t$ into the objective:

$$L(\theta) = \mathbb{E}_{t \sim U[0,1], (x_0, x_1) \sim \pi \mid x_t \sim W_{|x_0, x_1}^\epsilon} \left[\lambda(t) \left\| g_\theta(t, x_t) - (x_1 - x_0) - \sqrt{\frac{\epsilon t}{1 - t}} z \right\|^2 \right]$$

By multiplying the squared error by $1 - t$, the effective variance of the noise target becomes bounded (ϵt), ensuring stable gradients throughout the entire integration interval $[0, 1]$.

Crucially, under the proposed Gaussian mixture parameterization for the terminal potential, the drift $g_\theta(t, x)$ is not a black-box neural network requiring backpropagation through complex layers. Instead, it is evaluated directly using its exact analytical form:

$$g_\theta(t, x) = \epsilon \nabla_x \log \left\{ N(x ; 0, \epsilon(1 - t)\mathbf{I}) \sum_{k=1}^K [a_k N(r_k ; 0, \epsilon S_k) N(h_k(t, x) ; 0, A_k^t)] \right\}$$

where the quantities $h_k(t, x)$, and A_k^t are given by:

$$h_k(t, x) = \frac{1}{\epsilon(1 - t)}x + \frac{1}{\epsilon}S_k^{-1}r_k$$

$$A_k^t = \frac{t}{\epsilon(1 - t)}\mathbf{I} + \frac{1}{\epsilon}S_k^{-1}$$